GEEN-1171-M01-2023-24 Computer Aided Design and Manufacturing Assignment (University of Greenwich at Medway)

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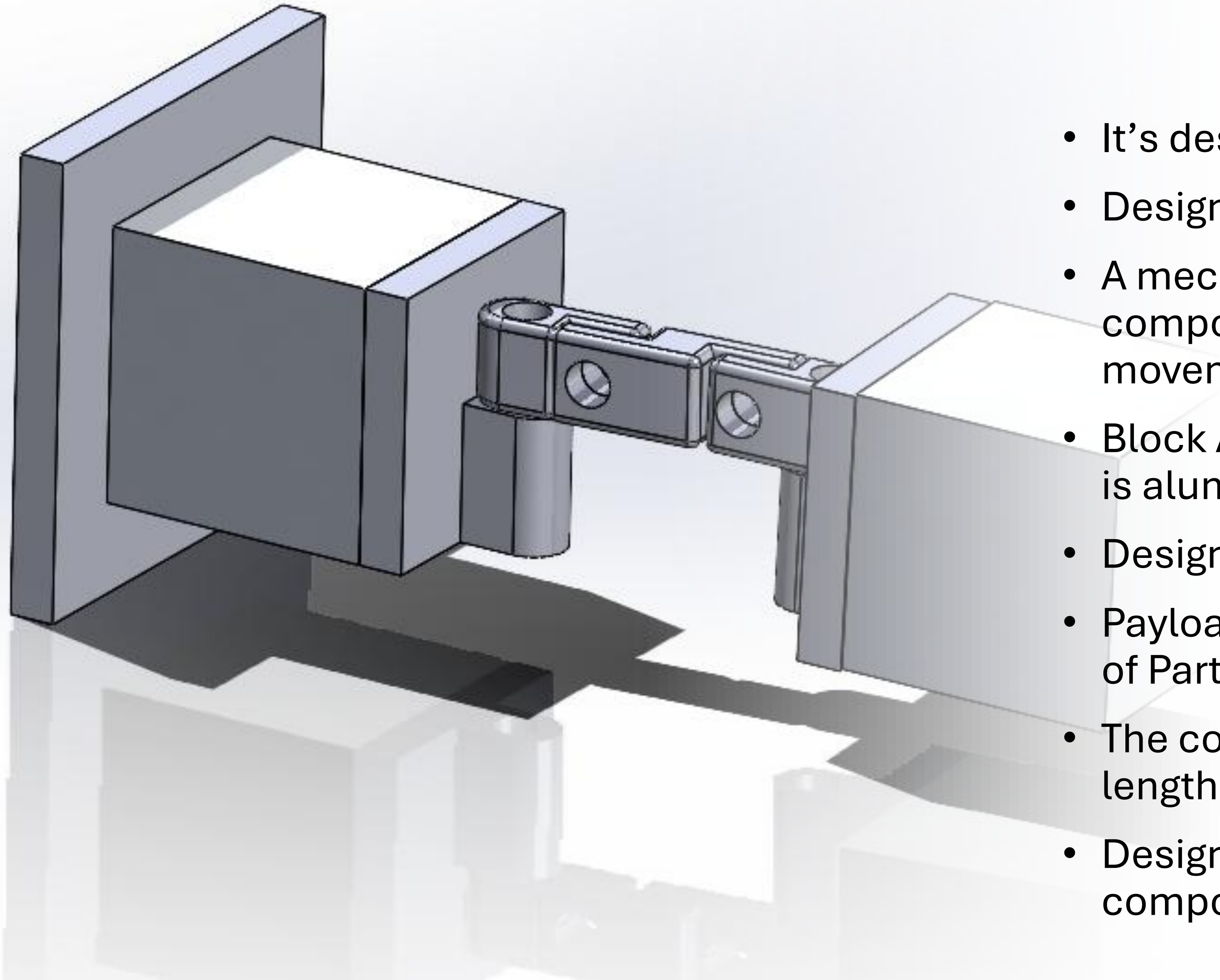
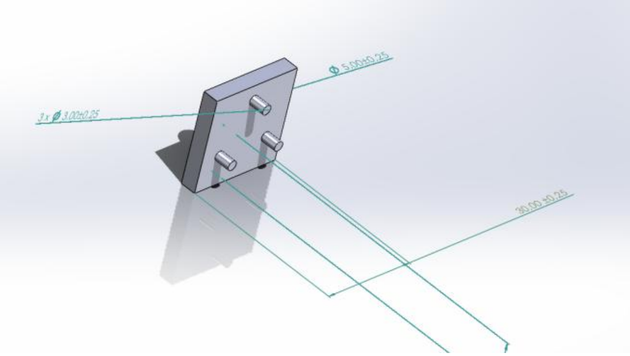
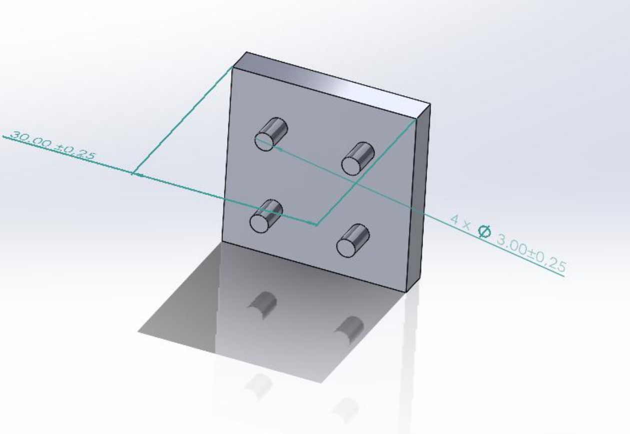
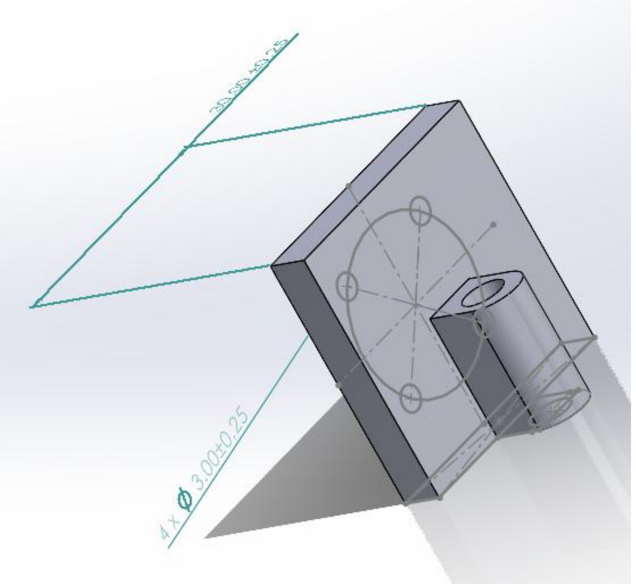
Specification of the Design in General

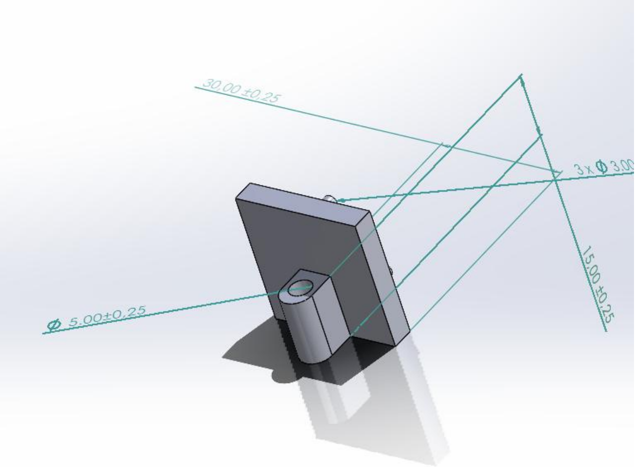


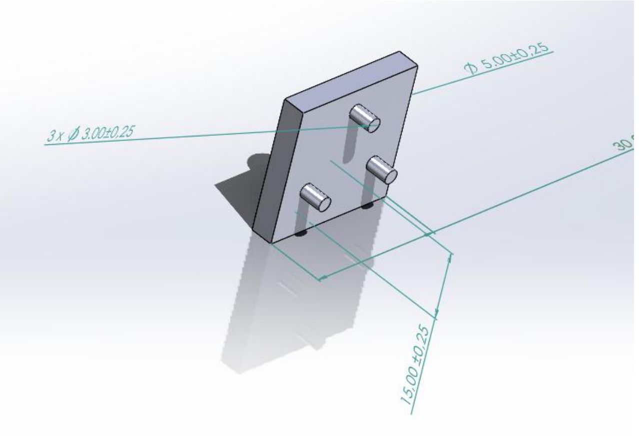
- It’s designed in in mm/Kg.
- Design mechanics (No Power).
- A mechanical connecting component permits 60 ˚ movement in the X/Y axis.
- Block A and Block B's material is aluminium 6061.
- Design is 60mm long.
- Payload capacity: 5 kg. (Mass of Part B)
- The connector's horizontal length is 60 mm.
- Design includes 7 main components.

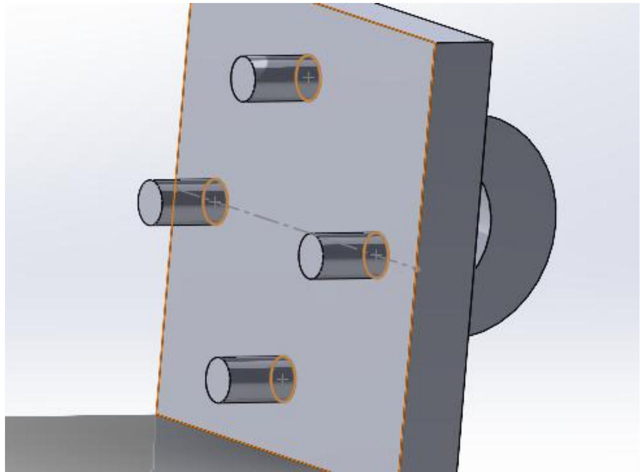








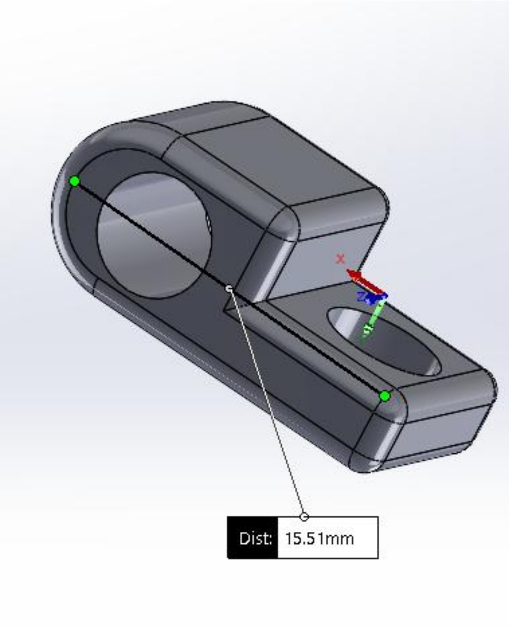


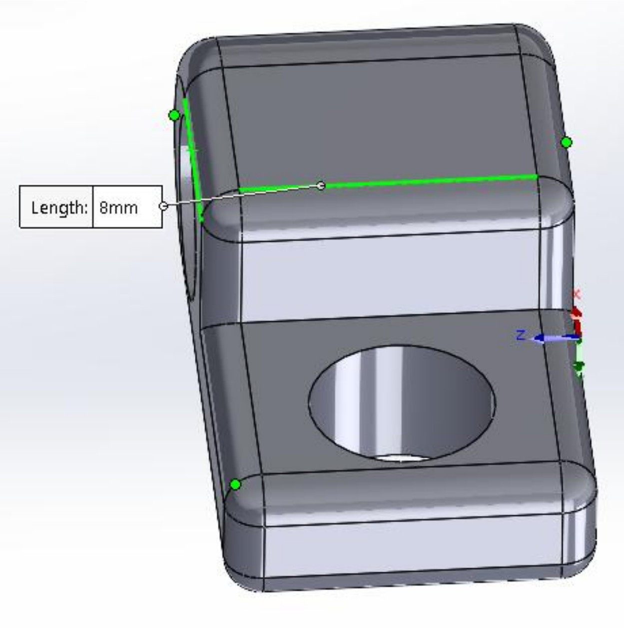


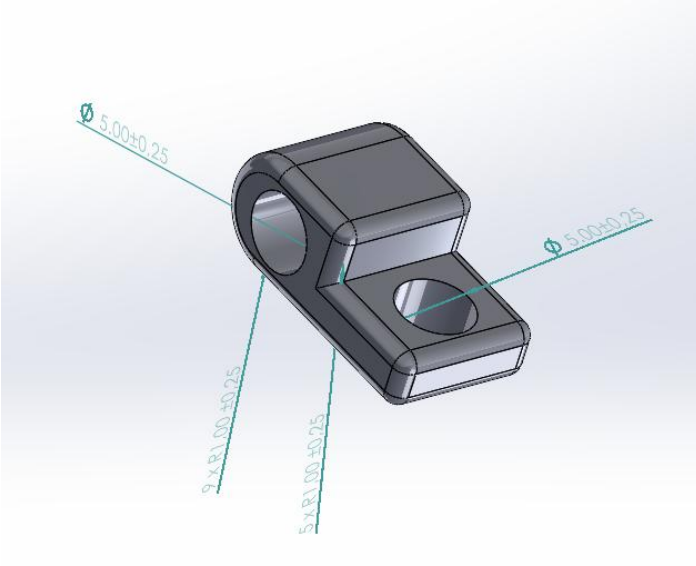
Specifications on components

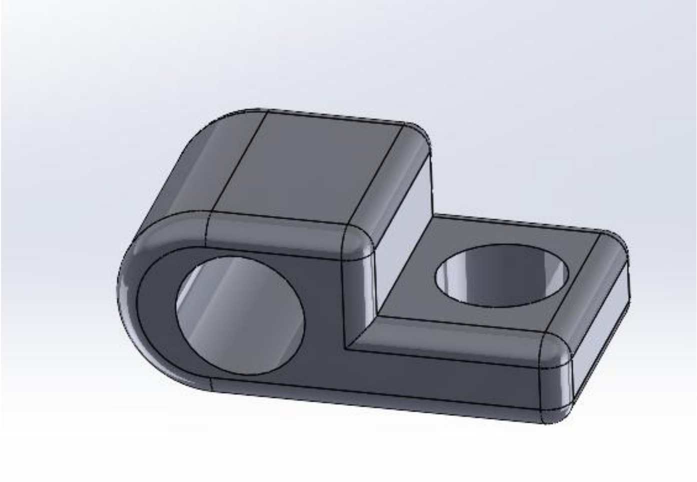
‘Part 1’

- Every tiny silindirs are properly designed for wholes in big blocks part A and B
- Replacement in that connection part which is in front side, looked more esthetic and it did not cause any problem in the design plan.
- The flat part which connect with blocks is 30mm-30mm.
- I had some problems about the places of teeths because it needs to fit with the blocks as you see in last photo in right column it did not fit in to that wholes in part A and B.
- Every teeths are 4x M3↓10 and placed equally into the flat area.



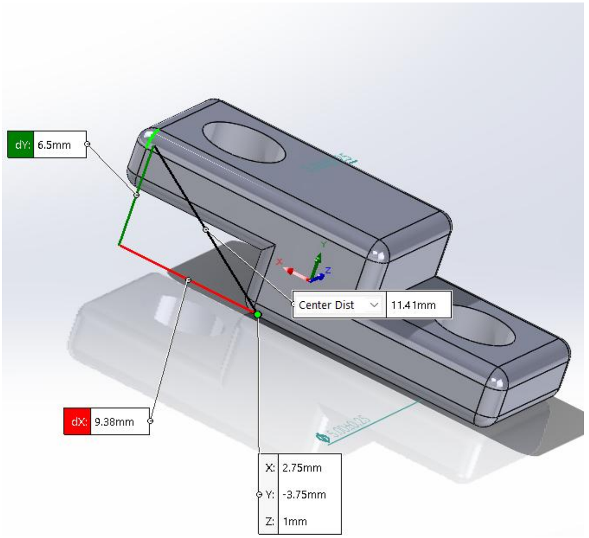


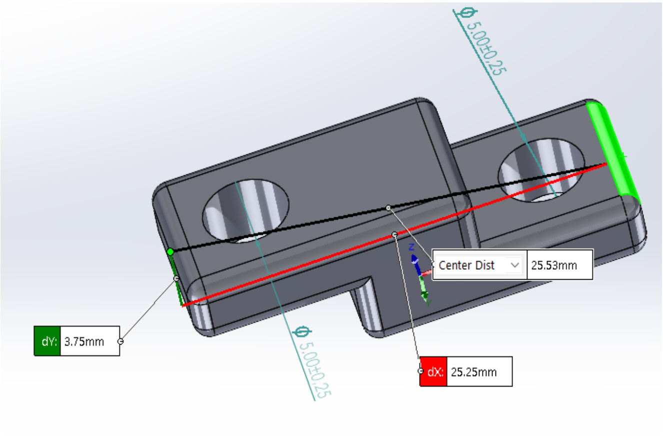


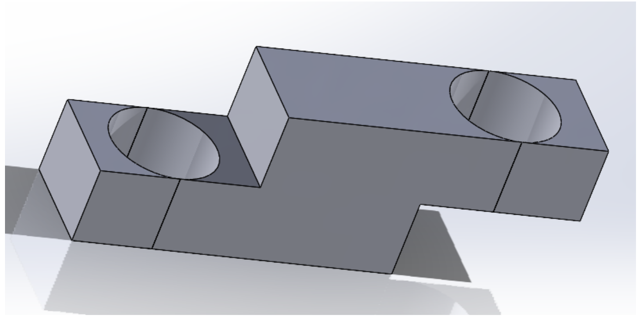


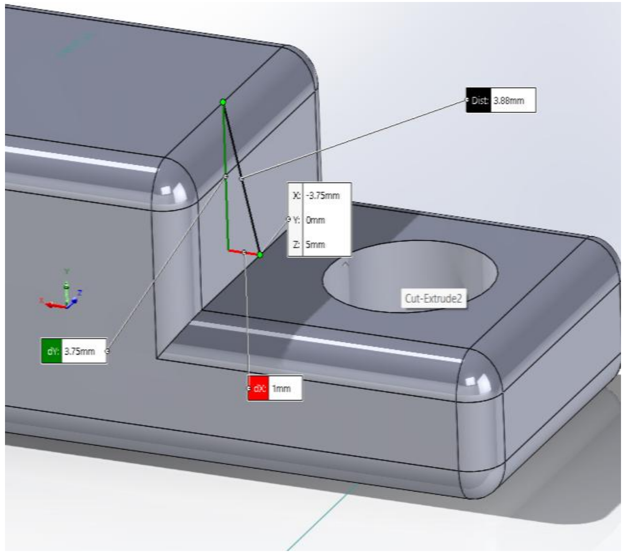
‘Part 2’

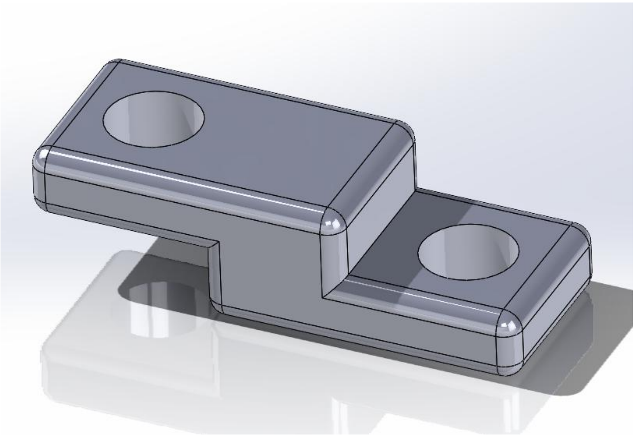
- The wholes which is designed for the pivot pins are equal and both of their radius are Ø5mm.
- The lenght is 17.5 mm long and width is 8mm.
- The edges in the design are properly designed to prevent crushing with other parts and also it looks estetically good.





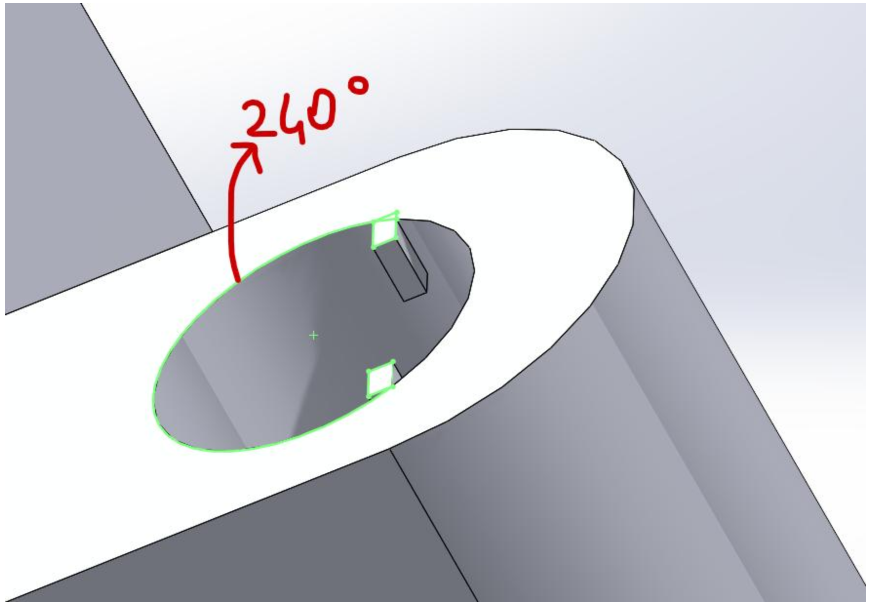


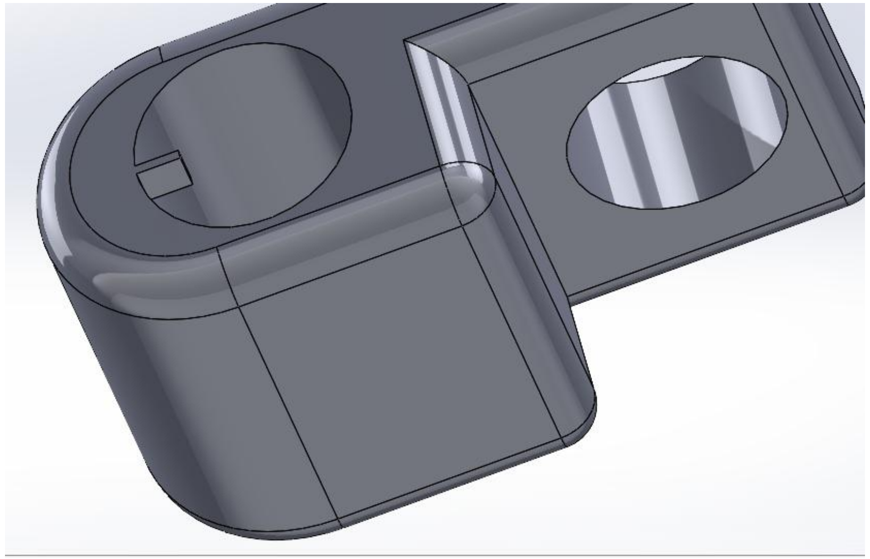




- **‘Part 3’**

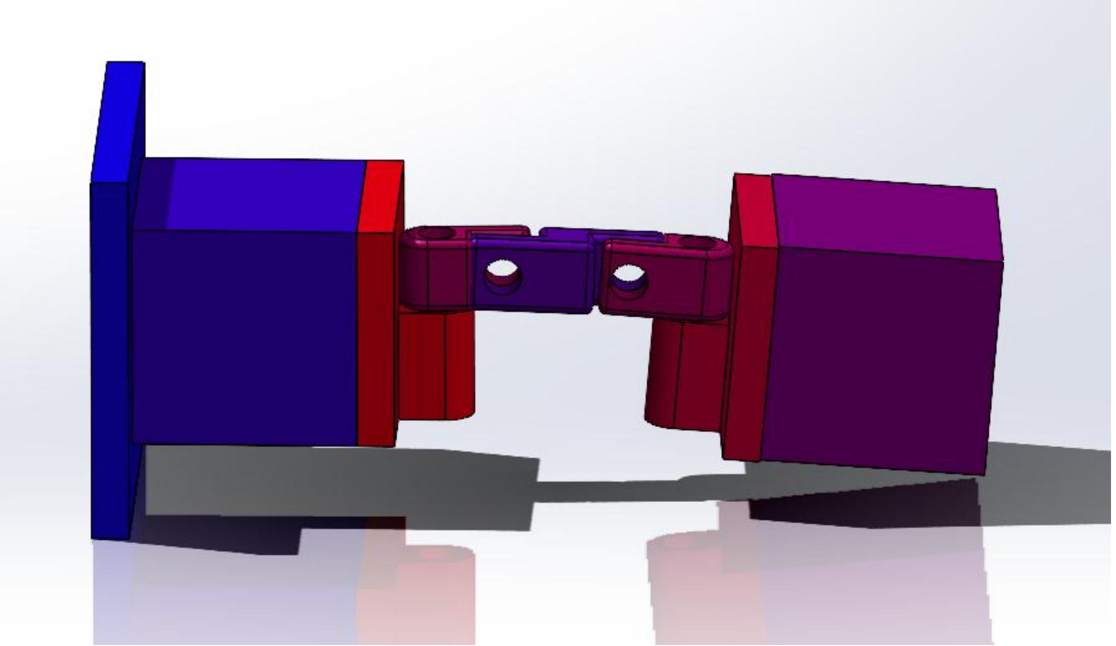
- Mid connector part is designed to attach the both part 2 components.
- The initial design has some open lines on the side, and I haven't created the edges, as you can see in the second photo in the right column.
  I created those lines for the same purpose as in Part 2's final product.
- Both wholes have sizes of Ø10mm and are evenly positioned in the middle of the planes.
- The lenght of component 3 is 25.25 mm, and its width is 6.5 mm. The depth is 3.75mm between the two sides.

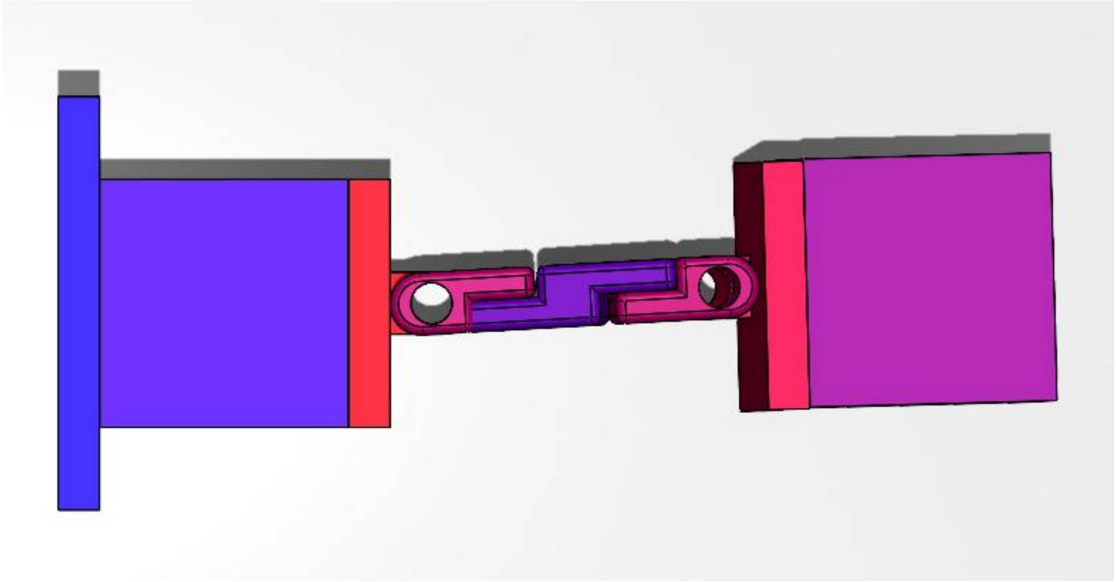




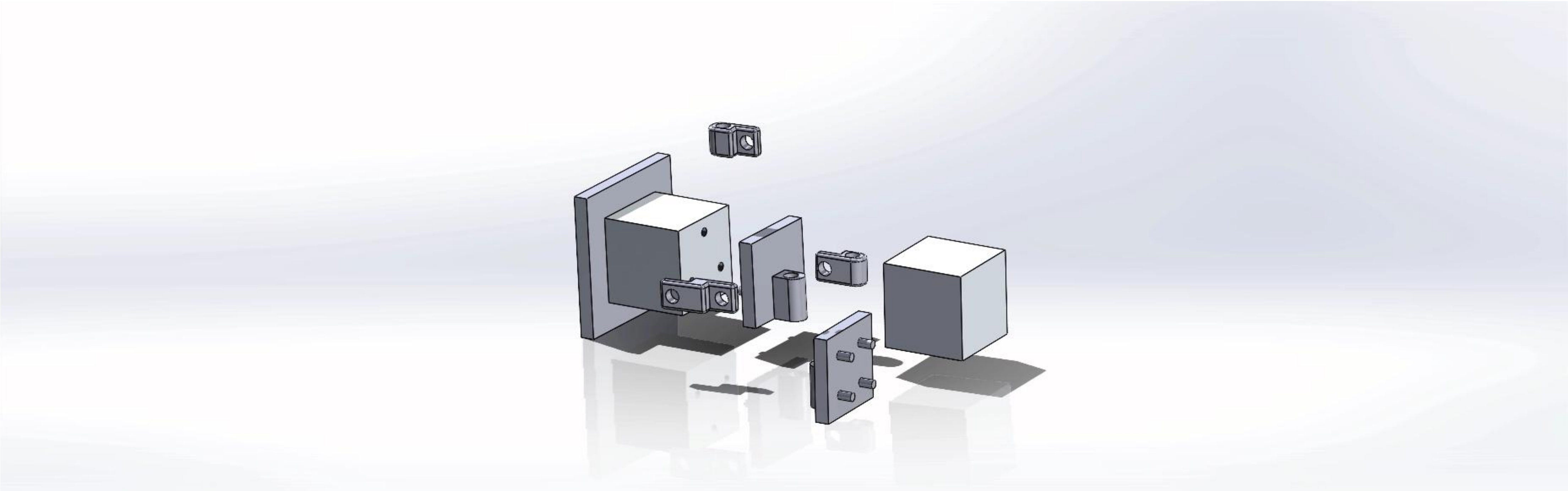
Range of motion

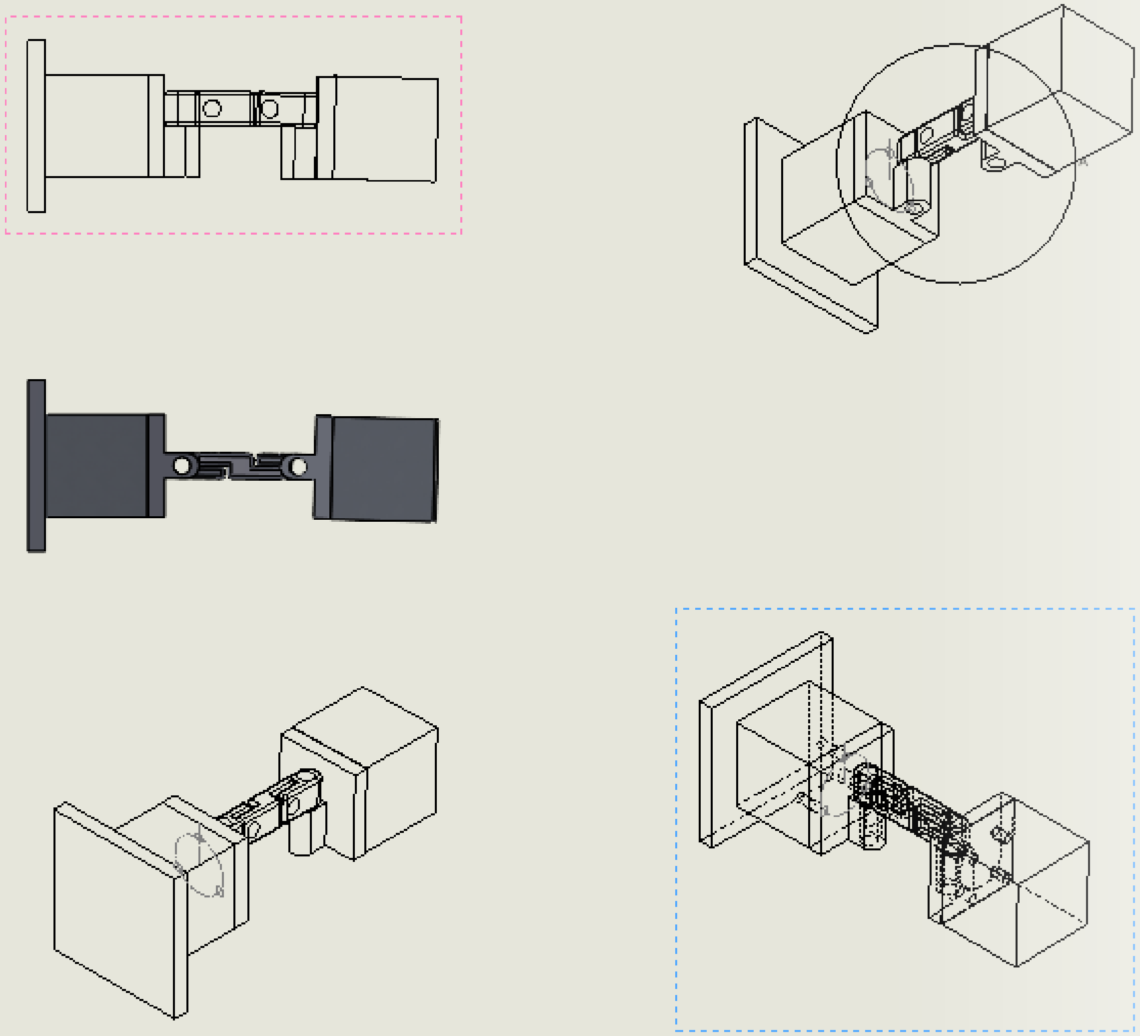
- We have 60 ˚movement in the both X/Y axis.
- My idea was adding pins to Part 1 and Part 2’s connection parts. In this way, pins crush each other and it allows the mechanical system to move only 60 ˚ on both axis.

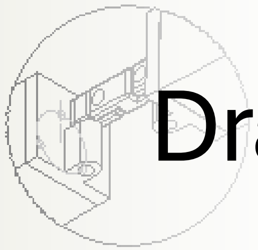




Assembly Animation

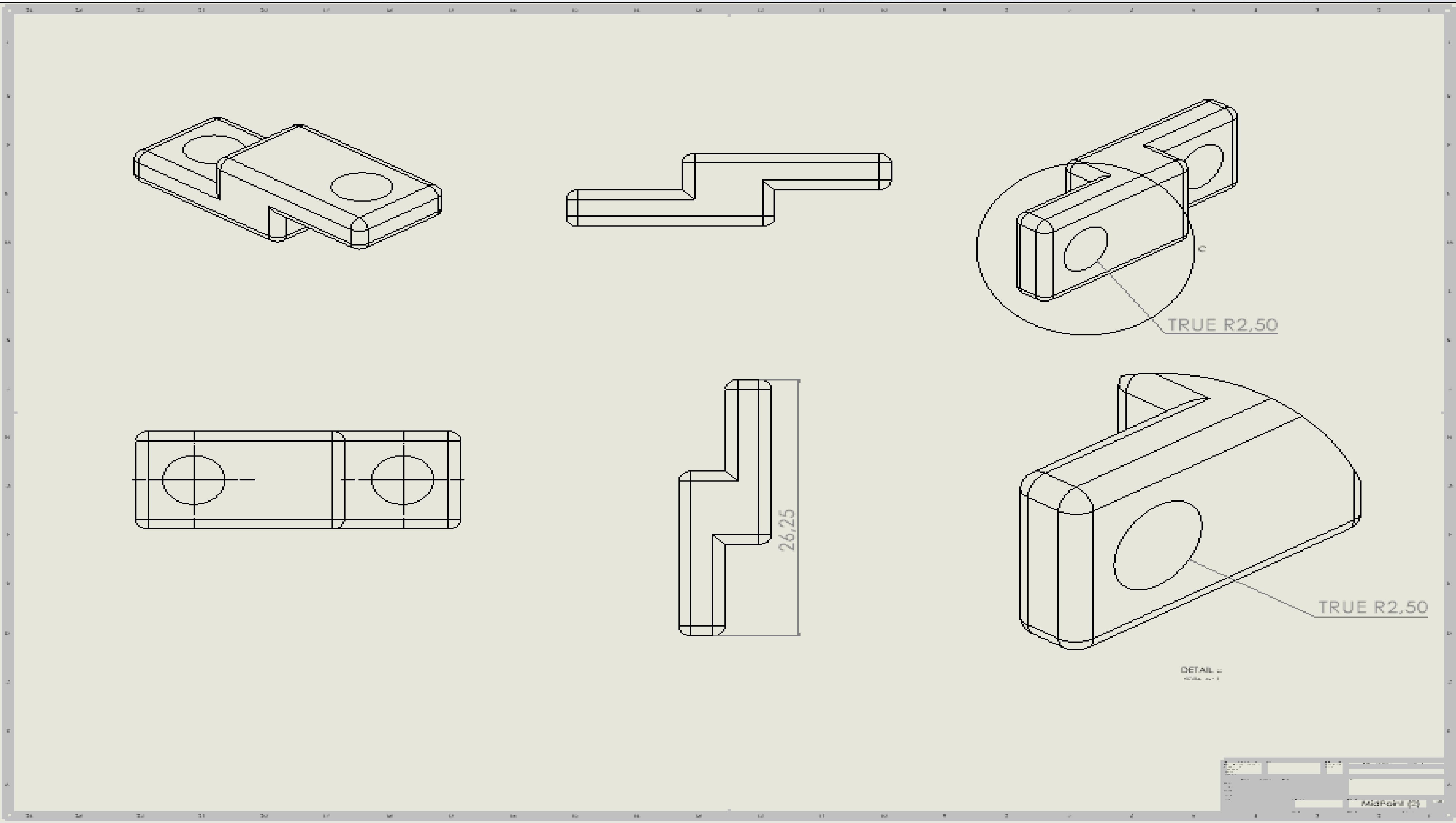




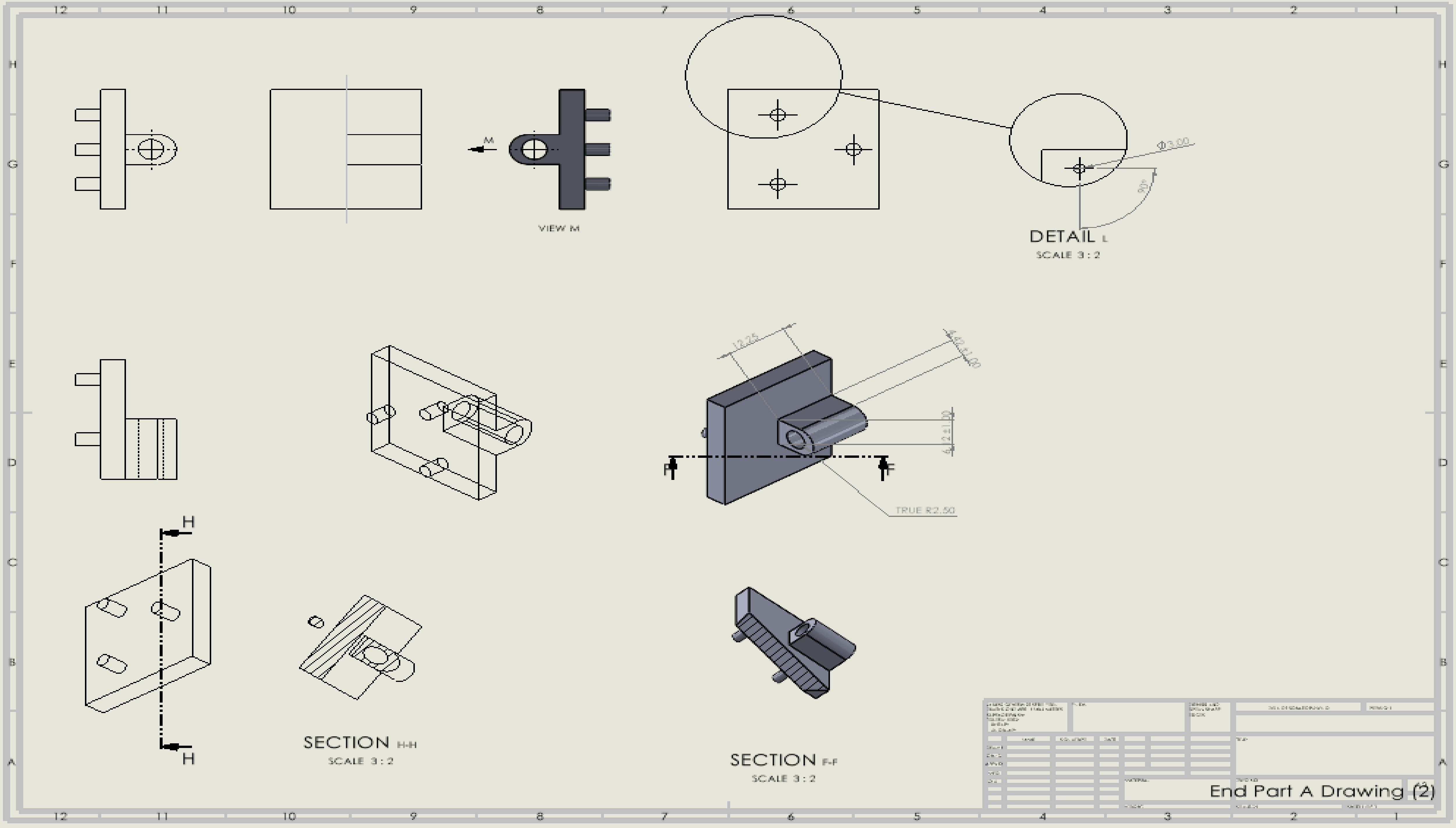


Drawings

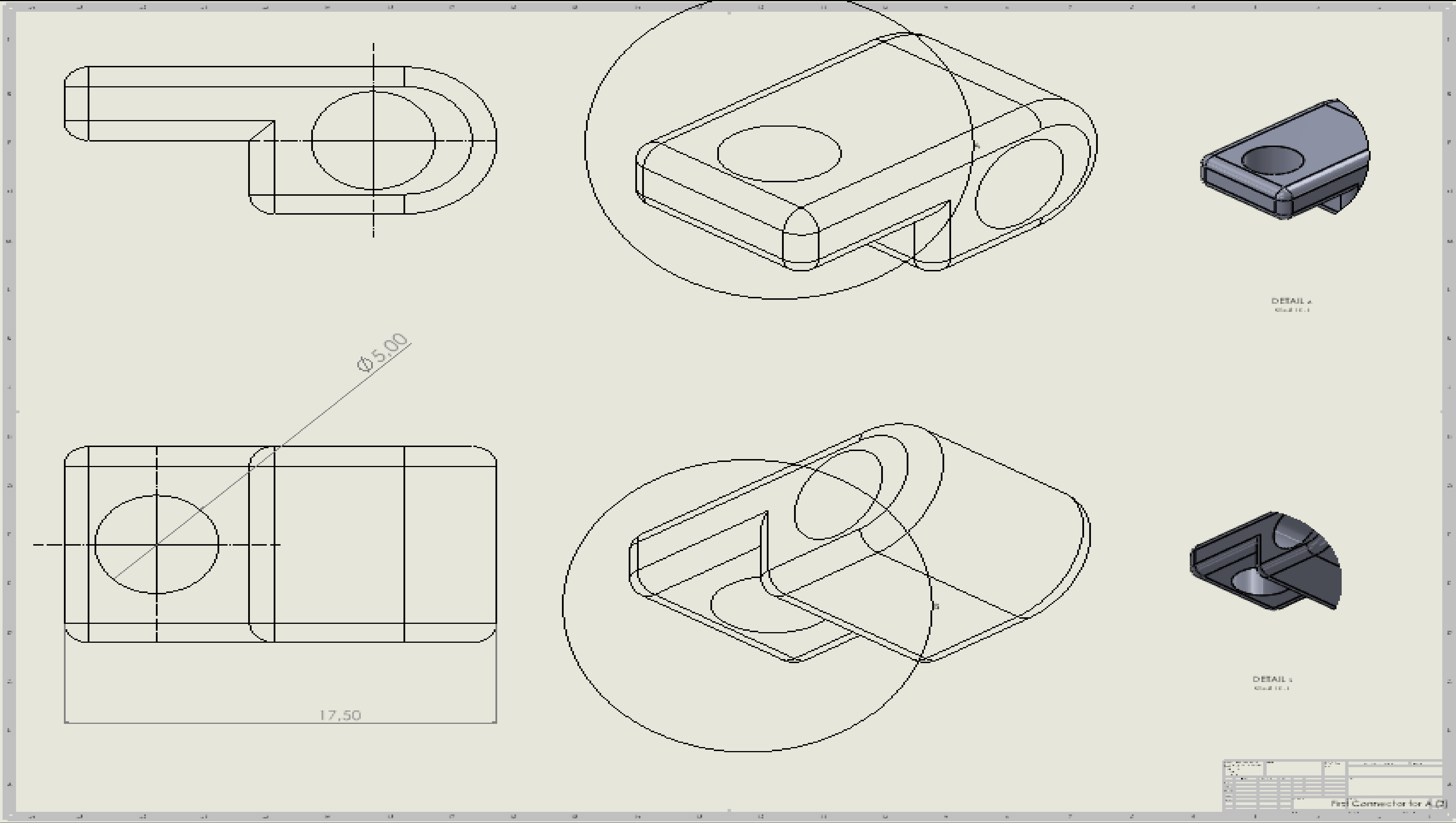
- I tried to convey certain proportions and the design's professional appearance in the illustrations.
- I have used the features of Detail View, Projected View, Auxiliary View, and so on to ensure you could clearly view their length in the part drawings.



TRUE R2,50
26,25
TRUE R2,50
DETAIL
MidPoint (2)



M
VIEW M
Φ3.00
90°
DETAIL L
SCALE 3 : 2
12.25
TRUE R2.50
F
F
H
H
SECTION H-H
SCALE 3 : 2
SECTION F-F
SCALE 3 : 2
End Part A Drawing (2)


Ø5,00
17,50
DETAIL
DETAIL
Connector for A (2)